

maxon

Internship project plan

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Explanation

This document outlines the project plan for my internship project.

The goal of the project is to implement an AI pipeline into a ctrlX App that will provide AI-based services and can be published in the ctrlX App Store.

The project is structured into sprints as in an Agile project management process. Sprints will have varying duration depending on the complexity of the tasks (typically 1-2 weeks).

Sprint	Focus Area	Main Output	Depends On
1	Discovery & Architecture	Architecture document + system setup	–
2	AI Refactoring	Deployable AI pipeline module	1
3	App Skeleton	Deployable ctrlX App template	1
4	Integration	AI pipeline connected to Data Layer	2, 3
5	UX & Interface	User interface + logging layer	4
6	Testing	Validated & secure application	4, 5
7	Deployment Prep	Store-ready snap package	6
8	Motor Interface	Added motor cycles	2
9	New Version	App from latest MIND	4
10	Testing and Documentation	Fully working application	All

Company Description

Maxon is a Swiss company headquartered in Sachseln, Switzerland, specializing in the development and production of high-precision electric drive systems. Their product portfolio includes brushless DC motors, planetary gearheads, encoders, and drive controllers, used in demanding fields such as medical technology, robotics, aerospace, and industrial automation.

This internship takes place within the Digital Technology department, which is responsible for developing the software systems that support and extend maxon's drive solutions. One of the key projects in this department is the MIND system, a predictive maintenance platform currently under active development. This internship contributes directly to that project by integrating the MIND AI pipeline into the ctrlX App ecosystem.

Internship details:

- Intern: Fanni Véh
- Company mentor: Dr Thomas Gittler (Project lead, CDTO)
- School supervisor: Jordi de Roeck
- Department: Digital Technology
- Location: Sachseln, Switzerland

Business Case

The MIND system is maxon's solution for predictive maintenance of electric motors. It tracks the condition of a motor throughout its lifecycle, continuously monitoring its state so that potential failures can be identified before they occur. This is especially critical for motors that must operate 24/7 without interruption - such as the motors used in hospital breathing machines or industrial production lines, where unexpected downtime can have serious consequences.

The system is currently at a stage where it is ready to be shared with clients for testing and feedback. To do this in a standardized and scalable way, the AI pipeline needs to be packaged as a ctrlX App, making it deployable directly on the controller and distributable through the ctrlX App Store.

This approach offers several advantages:

- Clients can install and run the MIND pipeline on their existing ctrlX hardware without additional infrastructure.
- Distribution through the App Store provides a controlled, validated channel for sharing the product with external users.
- Feedback from early adopters can be used to further develop the MIND system.

The project was conceived by maxon before the start of this internship. My role is to implement the integration of the pipeline into the ctrlX App architecture, bringing the product to a state where it can be published and tested externally.

Technology Stack

The project uses the following tools and technologies. Additional libraries may be added as the project progresses.

Technology	Purpose
Python	Core language; used by the existing MIND pipeline
SQLite	On-device database for storing motor and maintenance data
Alembic	Database schema and migration management
ctrlX App SDK	Framework for building and packaging ctrlX Apps
VS Code	Development environment
ctrlX Data Layer	Real-time data exchange with the controller
REST API	External access to AI inference results
snapcraft	Builds the deployable snap package for ctrlX

Sprint 1 - Setup & Discovery

This sprint focuses on understanding the existing system, setting up the development environment, and mapping out how the application will be structured inside the ctrlX ecosystem.

Tasks:

- Understand already existing MIND system
- Set up the controller-motor system with my own computer
- Install and configure the ctrlX App SDK and VS Code environment
- Explore the structure of a ctrlX App project (modules, services, snap confinement)
- Define system architecture and data flow
- Design reference trajectories and movements necessary for testing

Duration:

2 weeks

Sprint 2 - Refactoring

This sprint ensures that the AI pipeline is technically ready to be deployed into a ctrlX App.

Tasks:

- Refactor AI pipeline to be deployable
- Adapt the pipeline to the constraints of the ctrlX sandbox
- Optimize the pipeline for deployment

Duration:

1 week

Sprint 3 - Application setup

This sprint focuses on creating the base ctrlX App and establishing the development workflow from VS Code to the controller.

Tasks:

- Create base ctrlX App project using the App SDK template
- Understand how the generated project structure maps to runtime components
- Build the snap package and deploy the first version to the controller

Duration:

2 weeks

Sprint 4 - Integration

This sprint integrates the AI pipeline into the ctrlX App and connects all necessary data sources.

Tasks:

- Integrate the AI pipeline into the ctrlX application
 - AI module consumes live system data
 - AI inference results are written back to Data Layer
 - External systems can access results via defined APIs
- Connect the pipeline to the ctrlX Data Layer
- Implement communication between modules
- Add API endpoints

Duration:

2 weeks

Sprint 5 - UX & Interface

Once the application works and the AI pipeline is correctly implemented, the next sprint will focus on creating a usable and ctrlX-compliant interface.

Tasks:

- Align UX with ctrlX conventions
- Add user friendly extra features (error messages, logging)
- Ensure the UI follows ctrlX design guidelines

Duration:

2 weeks

Sprint 6 - Testing

This sprint ensures that the application behaves correctly inside the ctrlX sandbox and meets all functional requirements.

Tasks:

- Test the application across different scenarios
- Add necessary security measures
- Verify compliance with ctrlX App Store requirements

Duration:

1 week

Sprint 7 - Prepare for deployment

This sprint focuses on ensuring that the application is ready to be deployed and finishing any last moment changes that might be needed.

Tasks:

- Validate that application is ready for deployment
- Finalize the snap packaging and metadata
- Create user documentation, installation guide, etc.

Duration:

1 week

Sprint 8 – Motor Interface

The goal of this sprint is to close the known bugs and build the UI elements for motor cycle control.

Tasks:

- Fix the stop-cycle endpoint bug (causes internal error when stopping, preventing back-to-back cycles)
- Create interface button for starting/stopping motor cycles
- Add a live status bar showing current cycle state
- Investigate and resolve the import feature issue in the application

Duration:

1 week

Sprint 9 – New Version

Bring the newer MIND version up to full feature parity and resolve the prediction accuracy discrepancy.

Tasks:

- Port the motor data collection endpoints to MIND v1.2.3
- Add the cycle control interface elements to v1.2.3
- Talk to colleagues about why predictions behave differently on ctrlX vs. locally
- Verify that training and prediction work correctly on the ctrlX with v1.2.3

Duration:

1 week

Sprint 10 – Testing and Documentation

Ensure the application is stable, validated, and ready for the ctrlX App Store, as well as create thorough documentation for the application and all the changes.

Tasks:

- Test the application across different scenarios (good/bad cycles, edge cases)
- Verify ctrlX App Store compliance requirements
- Finalize snap packaging and metadata
- Create user documentation and installation guide

Duration:

1 week

Scope

In Scope:

- Refactoring AI pipeline for sandbox execution
- Integration into ctrlX App architecture
- Data Layer communication
- Defined REST API
- Basic UI aligned with ctrlX guidelines
- Snap packaging & metadata
- Functional testing
- Documentation

Out of Scope:

- Training new AI models
- Cloud-based AI services
- Large-scale distributed systems
- Advanced UI redesign beyond ctrlX standards
- Continuous deployment automation
- Performance optimization beyond required limits
- Hardware redesign
- Cybersecurity certification beyond store requirements

Risk Management

The following risks have been identified for this project.

Risk	Impact	Mitigation
ctrlX sandbox restricts Python dependencies or file system access	High	Sprint 2 is dedicated to adapting the pipeline to sandbox constraints before integration begins
SQLite or Alembic behaving unexpectedly within snap confinement	Medium	Database setup is tested early alongside the first app deployment in Sprint 3
Upstream changes to the MIND system affecting the integration	Medium	Clear interface boundaries are kept between the MIND pipeline and the ctrlX App layer
Unfamiliarity with the ctrlX App SDK causing unexpected delays	Medium	Sprint 1 includes dedicated time to explore and understand the SDK structure
Estimated 14-week plan cannot be completed in time	Low	The full internship is 17 weeks, leaving 3 additional weeks of buffer beyond the planned sprints

Stakeholders

The following people are involved in the MIND project and will be relevant stakeholders throughout this internship.

Name	Role
Gittler, Thomas	Project lead, CDTO (mentor)
Jaquemet, Claude	Product Owner
Geiser, Joel	Engineering lead, Software
Lorenz, Timo	Test-Bench, Hardware
Bacigalupo, Carlos	Data-Analytics
Joshi, Mihir	Data-Analytics
Carruzzo, Tristan	Data-Analytics

In addition, the end users of the MIND system are maxon clients in sectors where motors must operate continuously, such as medical device manufacturers and industrial automation companies.

Final Deliverable

The final result of this internship project is:

A production-ready ctrlX App containing an embedded AI inference pipeline that:

- Runs within ctrlX sandbox constraints
- Interfaces with ctrlX Data Layer
- Provides API endpoints for external systems
- Includes logging and UX alignment
- Can be published in the ctrlX App Store

Success Criteria:

- Successful deployment on controller
- Functional inference using live system data
- Stable runtime over defined test scenarios
- Store validation passed